

2920/206
DATABASE MANAGEMENT SYSTEM
July 2017
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION TECHNOLOGY

MODULE II

DATABASE MANAGEMENT SYSTEM

3 hours

INSTRUCTIONS TO CANDIDATES

You should have an answer booklet for this examination.

*This paper consists of **EIGHT** questions.*

*Answer any **FIVE** in the answer booklet provided.*

All questions carry equal marks.

Candidates should answer the questions in English.

This paper consists of 6 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1. (a) Define each of the following terms as used in databases (4 marks)
 - (i) Normalization;
 - (ii) Compound key.
- (b) Explain the term *database organization* as used in databases. (2 marks)
- (c) Distinguish between the role of a *database end user* and *database developer*. (4 marks)
- (d) (i) Table 1 is a structure of a table named *Department* in a database. Use it to answer the questions that follow.

Field	Description
DeptNo	Number(10)
DName	Varchar(20)
Location	Varchar(25)

Table 1

Write an SQL statement to perform each of the following: (8 marks)

- (I) Rename the field named *DeptNo* to *departmentName*;
 - (II) Remove the field named *Location*;
 - (III) Add a field named *dateStarted* having date type;
 - (IV) Remove the table.
 - (ii) Explain the data type *varchar* as used in the structure of the table (2 marks)
2. (a) Explain **two** reasons that may lead an organisation not to use a database management system. (4 marks)
 - (b) Outline **two** roles of each of the following in a database management system environment: (4 marks)
 - (i) Hardware;
 - (ii) Data.
 - (c) Interpret the following SQL statement: (3 marks)


```
SELECT deptNo, count(*)
FROM Employees
group by deptNo;
```

- (d) Table 2 shows unnormalised table. State a step by step process of normalising to 3NF. (9 marks)

Salesperson Number	SalesPerson Name	Customer Number	Customer Name	storeNumber	storeLocation	SaleAmount
2368	Jane	0005	Andrew	3	Hailesalase	3000
					Hailesalase	1500
					Moi Avenue	3000
2456	Ben	0009	Steven	5	Kijabe street	4500
3456	Gladys	0007	Maggie	7	Pangani	2500

Table2

3. (a) Outline **two** reasons for considering database security. (2 marks)
- (b) Explain **two** recovery techniques that can be used in case of a database failure. (4 marks)
- (c) Explain each of the following terms as used during the creation of tables in a database. (6 marks)
- (i) cascade;
 - (ii) restrict;
 - (iii) set null.
- (d) Study the following narrative and use it to answer the question that follows.
- A student must be enrolled on only one course and a course must have students enrolled on it. Each course is identified by a deptId, courseId and course name. Each course must have a number of modules and a module must be part of at least one course. A module is identified by moduleId, module name and must have a lecturer as a subject leader but a lecturer may not necessarily be a subject leader or may lead more than one module.*
- Draw an entity relationship diagram showing cardinality and optionality. (8 marks)
4. (a) Distinguish between the *Grant* and *Revoke* commands as used in databases. (4 marks)
- (b) With the aid of a diagram, distinguish between the output FROM the *union* and the *intersection* operators. (6 marks)
- (c) Explain **two** characteristics of the *conceptual model* of a database design. (4 marks)

- (d) (i) Write the algebraic expressions equivalent to the following SQL statements: (3 marks)

```
SELECT Distinct Gender, Salary
FROM Employee;
```

- (ii) Write the SQL statement equivalent to the following algebraic expression. (3 marks)

$$\sigma_{assetcategory=furniture}(Asset)$$

5. (a) Define the term *data abstraction* as used in database management systems. (2 marks)
- (b) Explain **two** disadvantages of database approach. (4 marks)
- (c) With the aid of an example, describe **two** categories of SQL statements as used in databases. (5 marks)
- (d) Table 3 shows a table named *Employee* in a database used by employer to manage employees' salaries. Use it to answer the question that follows

EmpNo.	EmpName	Salary	Grade
E234567	Catherine	10000	A
E234568	Agnes	15000	A
E234569	Alex	25000	D
E234570	Peter	25000	D
E234571	Ted	30000	C
E234573	Esther	75000	A

Table 3

Write an SQL statement that would: (9 marks)

- (i) display the salary of all employees whose Grade value is D.
- (ii) change Alex's salary to 30000.
- (iii) display all employees with grade A and salaries with values between 10,000 and 15,000.
- (iv) display the minimum and maximum salary.
6. (a) Outline **three** types of *end users* in a database environment. (3 marks)
- (b) Differentiate between the *prime attribute* and a *non-prime attribute* as used in databases. (4 marks)
- (c) Describe **two** ways of classifying database management system. (4 marks)

- (d) Table 4 shows a database table named students. Use it to answer the questions that follow:

SID	SNAME	LNAME	MARKS
001	Jane	Luke	50
002	Agness	James	70
003	Josephine	Eliud	90
004	Catherine	Joseph	40
005	Susan	Clark	30

Table 4

Write a relational calculus statement that would:

- (i) display the records for the fields *SID*, *SNAME* and *LNAME* of all students. (3 marks)
 - (ii) display the records in the field *SNAME*, *LNAME* and *MARKS* for all students with *MARKS* values more than 60. (3 marks)
 - (iii) display the *MARKS* values for a student with *SNAME* is "Catherine" and *LNAME* "Joseph". (3 marks)
7. (a) Explain each of the following terms as used in databases. (4 marks)
- (i) Two- tier client server architecture;
 - (ii) Candidate key.
- (b) Susan a database administrator is working on a table named *Employee* with the following attributes:
- Employee (EmployeeNumber, DepartmentNumber, hours, EmployeeName, DepartmentName, DepartmentLocation)*
- (i) State **three** functional dependencies that can be derived from the table. (3 marks)
 - (ii) Outline **three** challenges she would encounter when working with data on the table. (3 marks)
- (c) Explain **two** reasons that may lead to the migration from the file based to a database centred approach. (4 marks)
- (d) With the aid of a diagram outline the stages of the database design cycle. (6 marks)
8. (a) Explain **two** circumstances under which an organization would implement a centralized database system. (4 marks)
- (b) With the aid of a diagram, outline **two** paradigms the early models of databases were based on. (4 marks)

- (c) Distinguish between the entity relationship diagrams in figure 1 and in figure 2. (4 marks)

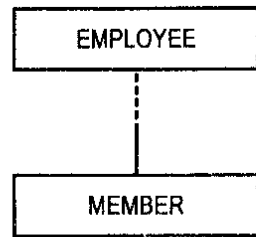


Figure 1

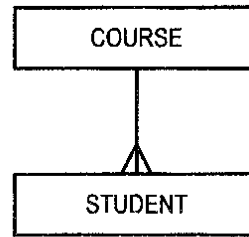


Figure 2

- (d) Write SQL statements that would perform each of the following operations on the table named *Department*: (8 marks)
- (i) Make the field *Dnumber*, a primary key;
 - (ii) Make the field *Dname* *VARCHAR(50)* not to accept a null value;
 - (iii) Make the field named *Dnumber* to hold values greater than 10

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